

# Calculus – Group Discussion

## Week 3 – Taylor and Maclaurin series

**ADSAI & CS / PR12**

## Learning goals (Week 3)

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- Identify critical points of functions.
- Analyse concavity using derivatives.
- Compute Taylor and Maclaurin series for elementary functions.
- Interpret Taylor polynomials as local approximations of functions.

## Critical points of functions

**Critical points.** A number  $c$  in the domain of  $f$  is called a *critical number* if

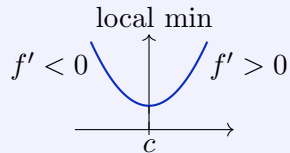
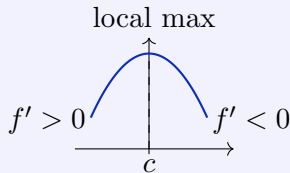
$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

The corresponding point  $(c, f(c))$  is called a *critical point* on the graph.

### First derivative test.

Suppose  $c$  is a critical number.

- If  $f'(x)$  changes from **positive** to **negative** at  $c$ , then  $f$  has a **local maximum** at  $c$ .
- If  $f'(x)$  changes from **negative** to **positive** at  $c$ , then  $f$  has a **local minimum** at  $c$ .
- If the sign of  $f'(x)$  does not change, then  $c$  is **not** a local extremum.



# Convexity and Concavity

**Concavity.** For a twice differentiable function  $f$ :

If  $f''(x) > 0$  on an interval, then  $f$  is **concave up** (or **convex**) there.

If  $f''(x) < 0$  on an interval, then  $f$  is **concave down** there.

A point where concavity changes is called an **inflection point**.

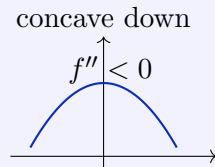
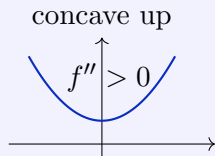
## Second derivative test.

Suppose  $f'(c) = 0$ .

If  $f''(c) > 0$ , then  $f$  has a **local minimum** at  $c$ .

If  $f''(c) < 0$ , then  $f$  has a **local maximum** at  $c$ .

If  $f''(c) = 0$ , the test is **inconclusive**.



## Linear approximations

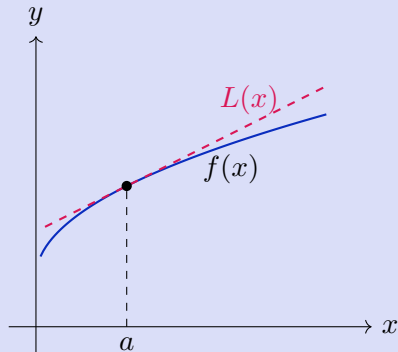
If  $f$  is differentiable at  $x = a$ , then near  $a$  the graph of  $f$  is well approximated by its tangent line. The **linear approximation** (or **linearization**) of  $f$  at  $a$  is

$$L(x) = f(a) + f'(a)(x - a).$$

**Idea.** For  $x$  close to  $a$ ,

$$f(x) \approx L(x) = f(a) + f'(a)(x - a).$$

This is useful for quick estimates of values that are hard to compute exactly.



Visualization: <https://www.desmos.com/calculator/jaecvccdpj?lang=nl>

# Taylor polynomials

The **Taylor polynomial of degree  $n$**  of  $f$  centered at  $x = a$  is

$$T_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x - a)^k$$

It matches the function and its first  $n$  derivatives at  $x = a$ .

## Why useful?

- local approximation;
- easier computation;
- basis for Taylor series.

$T_1(x)$  and  $T_2(x)$  of  $f(x) = \ln(x)$ :

<https://www.geogebra.org/m/rjttjrzn>

$T_{4-7}(x)$  of  $f(x) = \ln(x)$ :

<https://www.geogebra.org/m/axu72aah>

$T_{4-7}(x)$  of  $f(x) = \sin(x)$ :

<https://www.geogebra.org/m/wfez3qfw>

## Taylor series

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If the Taylor polynomials of  $f$  centered at  $a$  continue indefinitely, we obtain the **Taylor series**

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n.$$

When  $a = 0$ , this is the **Maclaurin series**.

### Interpretation.

- A Taylor series gives an infinite polynomial representation of a function near its center.
- In practice, we often use a **partial sum** to approximate the function.
- If a function is **analytic**, it can be defined by its Taylor series with perfect precision (no error)

## Key takeaways

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- Critical points come from  $f'(x) = 0$  or where  $f'(x)$  does not exist.
- The first derivative test classifies extrema via sign changes of  $f'$ .
- The second derivative describes concavity and helps classify critical points.
- Linear approximations use tangent lines to estimate function values locally.
- Taylor polynomials match derivatives at a center; Taylor series extend this idea to infinitely many terms.